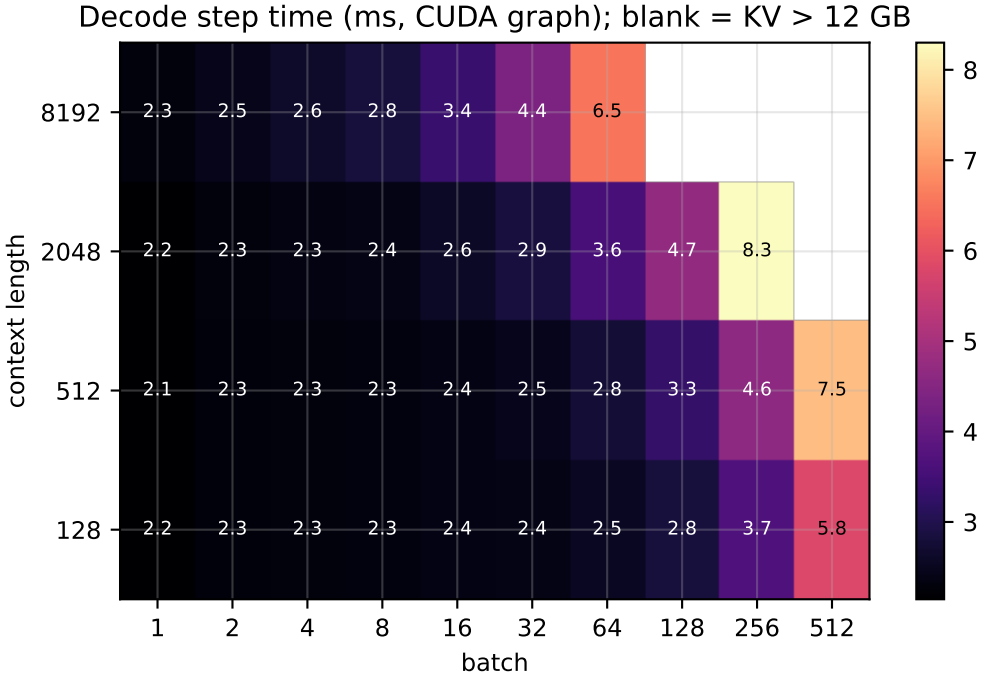
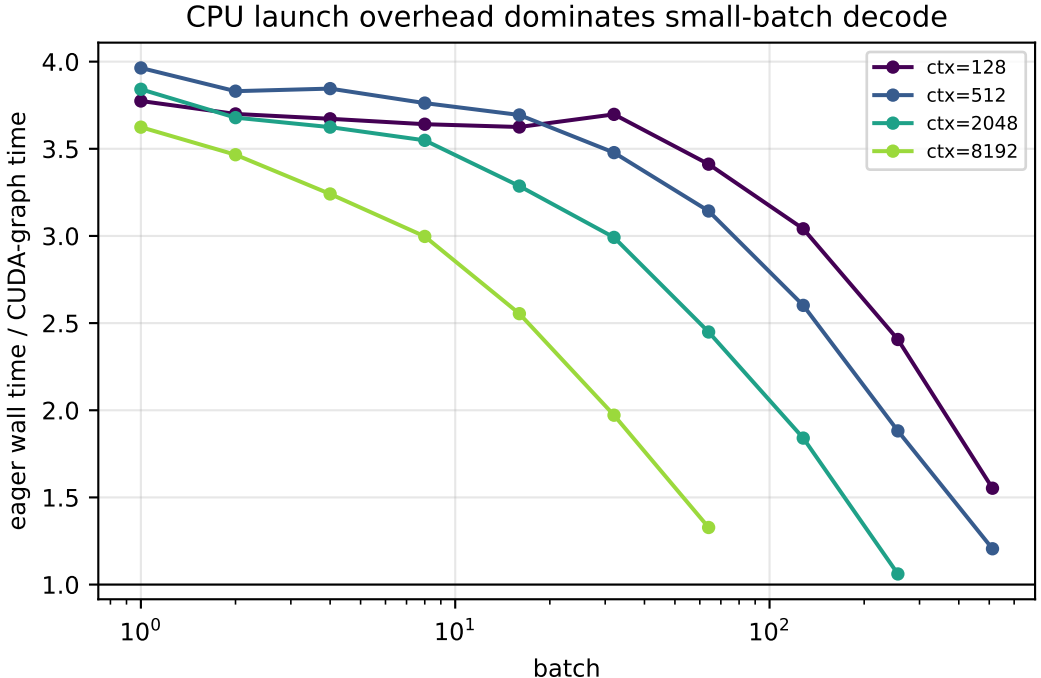

Decode step time (ms, CUDA graph); blank = KV > 12 GB
context length
8192
2048
512
128
2.3 2.5 2.6 2.8 3.4 4.4 6.5
2.2 2.3 2.3 2.4 2.6 2.9 3.6 4.7 8.3
2.1 2.3 2.3 2.3 2.4 2.5 2.8 3.3 4.6 7.5
2.2 2.3 2.3 2.3 2.4 2.4 2.5 2.8 3.7 5.8
1 2 4 8 16 32 64 128 256 512
batch
8
7
6
5
4
3


CPU launch overhead dominates small-batch decode
eager wall time / CUDA-graph time
4.0
3.5
3.0
2.5
2.0
1.5
1.0
10^0
10^1
10^2
batch
ctx=128
ctx=512
ctx=2048
ctx=8192